"HOMETOWN CARE FOR YOUR CHILD."

Medications for Attention Deficit Hyperactivity Disorder

There are several classes of medications that are commonly used for Attention Deficit Hyperactivity Disorder in children. This summary is not meant to be a class in pharmacology for parents but it hopefully will help to give a basic idea of the basic principles of how the most commonly used ADHD medications work and their potential side effects.

The Problem:

ADHD, both the hyperactive type as well as the inattentive type is primarily a deficit in attentiveness and impulse control. Even for individuals who are not at all hyperactive, they are impulsive in that they cannot keep from letting their consciousness wander, triggered either by external distractions (others talking, noise, etc.) or by thoughts and ideas within their head. For children with the hyperactive type of ADHD, the impulsiveness exhibits itself by inability to sit still, speaking out of turn and in more extreme cases, aggressive or defiant behavior.

The part of the brain that controls impulsivity/impulse control is the frontal lobe. When working well, the frontal lobe works as a filter—preventing blurting out of inappropriate language, allowing individuals to sit still, and block out noises and other distractions. How well the frontal lobe works correlates with the level of the neurotransmitter norepinephrine (NE). People who have *normal* NE levels in their frontal lobe tend not to have impulsivity/attention issues. Lower NE levels in the frontal lobe correlates with inattention and hyperactivity that are associated with ADHD. Increasing the frontal lobe's NE levels also correlates with improvement/resolution of these symptoms.

From a pharmacologic standpoint, medications target the NE level in the brain and specifically in the frontal lobe. Increasing the level tends to improve symptoms. To explain how to do this, we use the "bathtub" model. If the level of NE in the frontal lobe is modeled by a bathtub half full of water. The level of NE in the frontal lobe is represented by the level of water in the tub. There are two ways to increase the level of water—by turning on the spigot faster (i.e. by making more NE) or by obstructing the drain (i.e. by slowing down the rate that NE is broken down).

Stimulant Medication:

In general, stimulant medications (Ritalin, Adderall, Vyvanse, etc) cause neurons to fire off a bit more and thus result in the frontal lobe producing more NE. Although, each stimulant is slightly different and some may work better than others in a specific child, they all have more or less the same mechanism of action and the same potential side effects. Currently there is one medication, Strattera, that slows the rate of NE breakdown, thus increasing the level in the frontal lobe.

Advantages of stimulant medications are that they work quickly and are completely out of the system the same day. Thus, a patient will show immediate results when started on a stimulant. The dosage of these medications can be adjusted upward or downward fairly rapidly depending on level of improvement or side effects. Also, these medications are out of the patient's body the same day so any unwanted side effects that may be experienced are brief.

One of the downsides of stimulant medication is that they are "controlled substances" that are closely followed by the DEA. Prescriptions in California must be "hard copies" on paper. The medications cannot be faxed, called or electronically sent to pharmacies. Most insurance companies will only cover a 30-day supply of medication so at least initially parents need to pick up these prescriptions from the office and hand deliver them to the pharmacy to be filled. Once a patient is on a stable and effective dose of a medication, a 90-day prescription can be completed and sent by the parents to mail order pharmacies which most insurance companies have available.

As mentioned above, generally all stimulant medications have the potential for the same kind of side effects. Most of these are relatively mild and short lived, only occurring at the start of a medication trial or with an increase in dosage. Because the medications are 'stimulants,' one of the more common side effects is insomnia. When given in the morning, medications are usually out of the patient's system by dinnertime, so bedtime is not horribly affected. However in some sensitive individuals, there may be a few days initially where it may be hard to settle down to sleep. On occasion, in a child that is fidgety and has a hard time settling down at bedtime, this actually may improve with the start of medication. Another common side effect of stimulants is decreased appetite. These meds used to be prescribed for weight loss and so hunger at lunch time is often decreased. Patients on stimulants are followed closely for weight gain/loss. A child on stimulants should have a wholesome breakfast. Lunch is usually decreased, and the child should be offered a healthy snack after school and even before bed.

Other less common side effects include high blood pressure, agitation, muscle tremor, and anxiety. These usually only occur at medication doses higher than what are normally prescribed and so are not very likely at routine doses. One potential side effect of concern is motor tics (such as eye blinking, facial grimacing, throat clearing, etc.) The genes that encode for ADHD and tic disorders travel on the same chromosome. Very rarely, in younger children, starting on a stimulant medication may "unmask" or worsen the tendency to have tics. This usually will improve with a decrease or discontinuation of the medication, but on rare occasions the tics will persist. It is now felt that in these situations the development of motor tics was an eventuality that might have been hastened by use of stimulant but that the tics would have presented themselves regardless. The tendency to have motor tics runs in families and usually presents by about 4-6 years of age. If there is no family history and if the child is above age 6 and hasn't shown signs of tics than the risk is exceptionally small.

One other point to bring up is the abuse potential for stimulant medication. As mentioned earlier, these medications are controlled substances that require a "triplicate" prescription. Unfortunately, like many other prescription medications, stimulants can be stolen and used for recreational or non-prescribed purposes. The medications have "street value" and are commonly sold on high school or college campuses. It is not uncommon for me to hear stories of bottles of medications disappearing after workers have been in the house. That being said, stimulants and other controlled substances should be locked in an appropriate cabinet or lockbox to avoid this from happening.

Non-Stimulants:
As mentioned above, Strattera (atomoxetine) is the only currently approved medication that targets the rate of breakdown of NE in the brain. Benefits of this medication include that is **not** a controlled substance and thus does not have any abuse potential. Furthermore it is a medication that works 24/7—from the moment the patient rises until they go to bed. This makes the medication especially

useful for teenagers who are driving. Studies clearly show that individuals with ADHD who are not medicated are at increased risk for traffic accidents. Since most stimulant medications wear off by dinnertime, evening and night driving poses increased risk. However, because the medication has a long half-life, it takes about 3-4 weeks to take effect. In situations where students need to "salvage" the remainder of the school year, this is not the best choice. Oftentimes, patients can be transitioned from stimulants to Strattera over the summer when there can be a lag in effectiveness.

Potential side effects from Strattera are very mild and short lived. Initially there can be some stomach upset and drowsiness for a few hours after the dosage. This is remedied by giving Strattera with food in the evening. Generally within a week or two this resolves and the dose can be given any time of the day. The other downside of Strattera is that it only comes as a capsule. The powder inside the capsule is very bitter and doesn't mix well with foods or drinks. Thus, really only patients who can swallow pills are a candidate for this medication.

Other Medications:

Over the past few years several other classes of medications have been FDA approved for use in ADHD, either in combination with stimulants or by themselves. Intuiniv and Kapvay are really repackaged versions of older medications (guanfacine and clonidine) that have been used off label for decades to treat ADHD. Their mechanisms of actions are more complicated than the "bathtub" model. Often times adding these medications can prove helpful in adding to a stimulant regimen in a child who needs a bit more medication effect but doesn't tolerate the side effects of a higher dose of stimulants.

Summary:

One final point to make is that treating ADHD requires some patience from both the parent and the child. It often will take several attempts at different medications and different dosages before the right mix is found that helps to maximize ADHD treatment with a minimum of side effects. Unlike an ear infection where a standard antibiotic at a set dose will treat most every case successfully, ADHD treatment often takes several adjustments before an optimal treatment regimen is found. Follow-up phone calls and office visits are needed to monitor progress and track growth. Good communication between patient/parent and the doctor's office will help result in a successful treatment program.

--Be Well

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